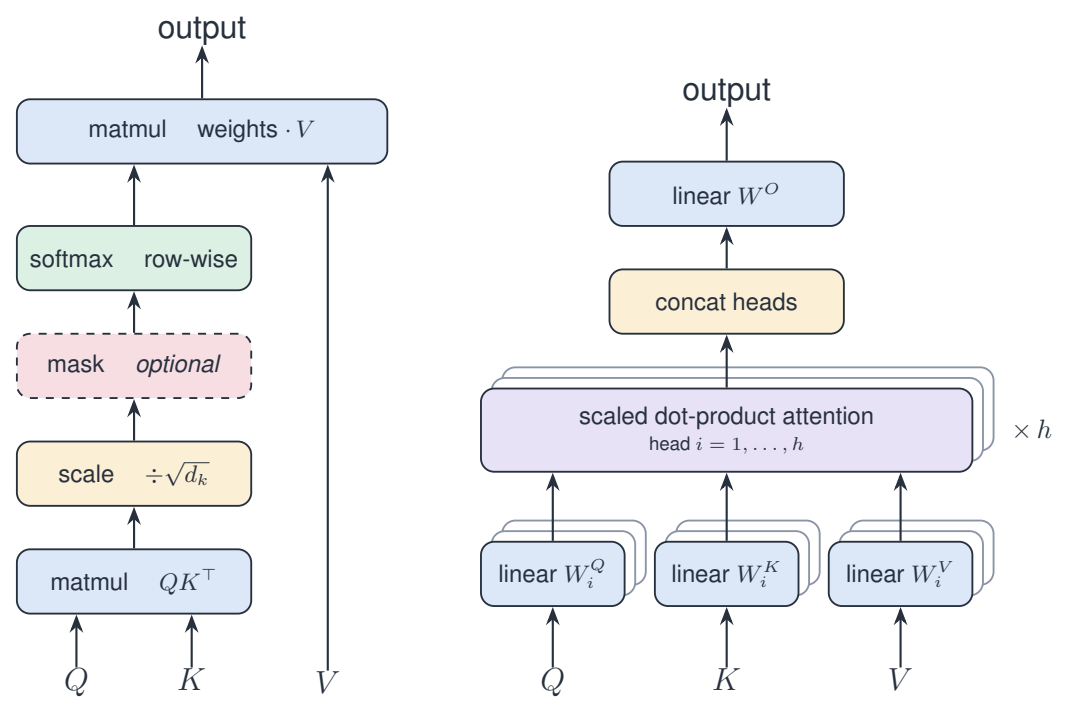

output
matmul weights · V
softmax row-wise
mask optional
scale ÷√d_k
matmul QK^⊤
Q
K
V
output
linear W^O
concat heads
scaled dot-product attention
head i = 1, . . . , h
× h
linear W_i^Q
linear W_i^K
linear W_i^V
Q
K
V